

PolyArt: Customizable Multilingual Movie Poster Generation via Diffusion Transformer

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Figure 1: Posters results generated by PolyArt. The backgrounds, layouts, and typography designs are directly generated from the text input, demonstrating that PolyArt can produce posters with high textual accuracy and aesthetic appeal. Moreover, the poster text can be customized in style and content, supporting four languages: English, Chinese, Korean, and Japanese.

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1 Introduction

Movie posters are key to audience engagement and core messaging. Yet, current AI generation methods face limitations. Base diffusion models [Black Forest Labs 2024; Stability AI 2024] struggle to produce aesthetic posters with accurate text. End-to-end methods [Chen et al. 2025a] generate complete posters in one step but lack fine-tuning capability. Furthermore, other specialized generation methods [Chen et al. 2025b] only support English. We propose PolyArt, a multilingual aesthetic poster framework. Its core contributions include: (1) the DeArtifact Cinematic LoRA, achieving high-fidelity, cinematic visual rendering rich in aesthetic details;

(2) the Custom-specified Text Refiner, supporting fine-grained customization of text content and style in Chinese, English, Japanese, and Korean. As shown in Figure 1, PolyArt generates aesthetic details and enables fine-grained control over poster text.

2 Method

Our PolyArt method advances the diffusion transformer (DiT) model with two key innovations, including the DeArtifact Cinematic LoRA (DC-LoRA) and the Custom-specified Text Refiner (CTR), as shown in Figure 2. During the Enhanced Generation Phase, the DC-LoRA is integrated with the base model to synthesize aesthetically enhanced posters. In the Interactive Customization Phase, the generated poster undergoes evaluation. When refinement is required, the CTR module is activated, enabling modifications to the text appearance and content on the poster through textual style and content conditions. Compared to other methods, PolyArt generates posters with outstanding aesthetic quality and supports four languages.

Enhanced Generation Phase. While text-driven poster base image generation frameworks can ensure a certain level of textual accuracy and output basic aesthetic posters, they still exhibit limitations in expressiveness when handling textural details and visual richness. To overcome this constraint, we trained DC-LoRA [DiffSynth-Studio 2025] on the IMDb dataset [IMDb 2024]. This dataset comprises 9,730 authentic movie poster images, each measuring 182 by 268 pixels. Each poster image in the dataset is paired with its corresponding background description. The training process utilized

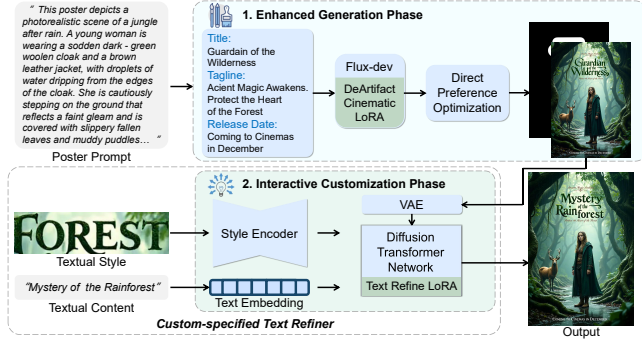


Figure 2: Our PolyArt pipeline consists of two phases. (1) Enhanced Generation Phase: produces high-quality posters. (2) Interactive Customization Phase: optimizes poster text based on textual style (image) and textual content (text).

8 NVIDIA A100 GPUs for 100 epochs with a learning rate of $1e-4$, requiring approximately three hours to complete. Then, the trained DC-LoRA is injected via low-rank matrices into the image and text attention layers, multilayer perceptron layers, linear layers, and modulated linear layers of the base model. This guides the model to generate visual structures that better conform to human aesthetic preferences based on the poster prompt.

Interactive Customization Phase. If the text on posters generated by the Enhanced Generation Phase requires modification or style optimization, the CTR module activates to enable fine-grained control. Given a textual style image, the style encoder extracts fine-grained style features from it. The style encoder integrates a pre-trained multi-modal visual encoder [Zhai et al. 2023] with a secondary module structured as linear layers followed by a Qformer [Li et al. 2023]. Text style transfer is then achieved in the latent space by acting on the key-value pairs in the attention layers of the denoising network. When provided with textual content, CTR generates multilingual text by loading Text Refine LoRA [Xie et al. 2025] in the DiT during the sampling process. Finally, the textual style image is concatenated with the masked input. Through the self-attention mechanism of the diffusion model, the textual content seamlessly blends with the background and unmodified text [Ma et al. 2025].

3 Experiment and results

We use DeepSeek [DeepSeek AI 2024] to randomly generate 100 aesthetic poster prompts for qualitative and quantitative comparisons across five leading methods. We also conduct ablation studies to validate the effectiveness of DC-LoRA. More details are presented in the supplementary material.

In qualitative comparisons, as shown in Figure 3, Playground-v2.5 [Li et al. 2024] and SD3.5 [Stability AI 2024] struggle to handle long text. Flux1.dev [Black Forest Labs 2024] and BAGEL [Deng et al. 2025] frequently generate unrecognizable or misspelled text. The PosterCraft [Chen et al. 2025a] method generates posters in a single step, lacking tunability. All five methods face challenges in correctly processing non-English text. Our method significantly outperforms others in terms of multilingual text rendering accuracy, aesthetic appeal, prompt alignment, and customizability.



Figure 3: Qualitative comparisons of different methods' outputs. Red boxes highlight misspelled text, while yellow boxes indicate redundant or missing text elements.

Table 1: Quantitative comparisons of different model outputs.

Method	Accuracy↑	Recall↑	F1↑
Playground-v2.5 [Li et al. 2024]	0.044	0.019	0.024
SD3.5 [Stability AI 2024]	0.499	0.452	0.445
Flux1.dev [Black Forest Labs 2024]	0.538	0.479	0.472
BAGEL [Deng et al. 2025]	0.470	0.517	0.476
PosterCraft [Chen et al. 2025a]	0.629	0.607	0.591
Ours	0.682	0.709	0.676

In the quantitative comparison experiment, ten users with AI tool experience and artistic backgrounds evaluated four key aspects: aesthetic appeal, text readability, prompt alignment, and overall effect, using a scale of 1 to 10. PolyArt demonstrates leading performance in aesthetics, text readability, and prompt alignment. More details are presented in the supplementary material. Additionally, for textual accuracy, we employ an OCR tool [Wei et al. 2024] to recognize text in generated posters and compute overall character-level precision, recall, and F1-score using micro-averaging. As shown in Table 1, our method achieves the best quantitative results.

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